

Suren Network | Learning platform | PNETLab | Topology | what is the command for tty lin

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Notifications

R1

```
Gateway of last resort is 20.0.0.2 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 20.0.0.2
C 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 10.0.0.0/8 is directly connected, Ethernet0/0
L 10.0.0.1/32 is directly connected, Ethernet0/0
C 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/1
L 20.0.0.1/32 is directly connected, Ethernet0/1
S 30.0.0.0/8 [1/0] via 20.0.0.2
S 40.0.0.0/8 [1/0] via 20.0.0.2
S 172.50.0.0/16 [1/0] via 20.0.0.2
S 200.12.56.0/24 [1/0] via 20.0.0.2
R1#
*Jun 20 06:40:11.212: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1#
```

Network topology diagram showing a central router R3 connected to three other routers: R1, R2, and R4. R1 is connected to Win-1 and Win-2. R2 is connected to Linux-1 and Linux-2. R4 is connected to DNS. The diagram shows various IP addresses and interfaces for each device.

ACL

Physical

01 : 00 : 00 START

Hot weather Now

Search

ENG IN

12:32 20-06-2026

Screenshot of a PNETLab topology simulation interface showing a network diagram and a terminal window for router R2.

Terminal Window (R2):

```
Gateway of last resort is not set
S 10.0.0.0/8 [1/0] via 20.0.0.1
C 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 20.0.0.0/8 is directly connected, Ethernet0/0
L 20.0.0.2/32 is directly connected, Ethernet0/0
C 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.1/32 is directly connected, Ethernet0/1
C 40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 40.0.0.0/8 is directly connected, Ethernet0/2
L 40.0.0.1/32 is directly connected, Ethernet0/2
S 172.50.0.0/16 [1/0] via 30.0.0.2
S 200.12.56.0/24 [1/0] via 40.0.0.2
R2#
```

Network Diagram:

- Left Network (10.0.0.0/8):** Contains two PCs (Win-1, Win-2) connected to a switch (Net) with IP 10.0.0.1. The switch is connected to router R1 (20.0.0.1) via Ethernet0/0. R1 is connected to router R2 (20.0.0.2) via Ethernet0/1.
- Central Router (R2):** Configured with interfaces e0/0 (20.0.0.2), e0/1 (30.0.0.1), and e0/2 (40.0.0.1).
- Top Right Network (172.50.0.0/16):** Connected to R2 via e0/1 (30.0.0.2). Contains a switch (Net) with IP 172.50.0.1, which is connected to two Linux servers (Linux-1, Linux-2) with IP addresses 172.50.0.10 and 172.50.0.20.
- Bottom Right Network (200.12.56.0/24):** Connected to R2 via e0/2 (40.0.0.2). Contains a switch (Net) with IP 200.12.56.1, which is connected to a DNS server (DNS) with IP address 200.12.56.10.

Interface Summary:

Router	Interface	IP Address
R1	Ethernet0/0	10.0.0.1
R1	Ethernet0/1	20.0.0.1
R2	Ethernet0/0	20.0.0.2
R2	Ethernet0/1	30.0.0.1
R2	Ethernet0/2	40.0.0.1
Switch (Top Right)	Net	172.50.0.1
Switch (Bottom Right)	Net	200.12.56.1

Bottom Bar: Includes a search bar, system tray with date/time (12:32, 20-06-2026), and a "START" button.

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```
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is 30.0.0.1 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 30.0.0.1
C   30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     30.0.0.0/8 is directly connected, Ethernet0/1
L     30.0.0.2/32 is directly connected, Ethernet0/1
L   172.50.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     172.50.0.0/16 is directly connected, Ethernet0/0
L     172.50.0.1/32 is directly connected, Ethernet0/0
R3#
*Jun 20 06:46:37.650: %PNF-6-PNF_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R3#
```

Win 2
10.0.0.20

30.0.0.2
e0/1
e0/0
172.50.0.1
Net
Linux-1
172.50.0.10
Linux-2
172.50.0.20

0.0.1
e0/1
e0/2
40.0.0.1
40.0.0.2
200.12.56.1
R4
e0/0
Net
DNS
200.12.56.10

ACL

Physical

01 : 00 : 00 START

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Notifications

R4

```
+ - replicated route, % - next hop override, p - overrides from Pfr
Gateway of last resort is 40.0.0.1 to network 0.0.0.0

R4#
R4# 0.0.0.0/0 [1/0] via 40.0.0.1
R4# C 40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
R4# L 40.0.0.0/8 is directly connected, Ethernet0/1
R4# L 40.0.0.2/32 is directly connected, Ethernet0/1
R4# C 200.12.56.0/24 is variably subnetted, 2 subnets, 2 masks
R4# L 200.12.56.0/24 is directly connected, Ethernet0/0
R4# L 200.12.56.1/32 is directly connected, Ethernet0/0

*Jun 20 06:46:53.781: $PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R4#
```

Win-1
10.0.0.10

Win-2
10.0.0.20

Net

10.0.0.1

20.0.0.1

20.0.0.2

30.0.0.1

40.0.0.1

40.0.0.2

200.12.56.1

172.50.0.1

172.50.0.10

172.50.0.20

Linux-1

Linux-2

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

ACL

Physical

Hot weather Now

Search

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20-06-2026

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Notifications

QEMU (Linux-1)

```
ifconfig eth0 Link encap:Ethernet HWaddr 50:60:F6:02:66:80  
inet addr:172.50.0.10 Bcast:172.50.255.255 Mask:255.255.0.0  
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1  
RX packets:185 errors:0 dropped:0 overruns:0 frame:0  
TX packets:160 errors:0 dropped:0 overruns:0 carrier:0  
collisions:0 txqueuelen:1000  
RX bytes:58517 (57.1 KiB) TX bytes:57960 (56.6 KiB)
```

10.0.0.10
10.0.0.20

172.50.0.1
172.50.0.10
172.50.0.20

40.0.0.2
200.12.66.1
200.12.66.10

ACL

Physical

01 : 00 : 00 START

Rain coming
In about 3 hours

Search

ENG IN

12:33
20-06-2026

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Notifications

QEMU (Linux-2)

```
route: ioctl 0x030c failed: No such process
Removing default gateway for Ethernet interface ..... [FAILED]
Shutting down Ethernet interface eth0 ..... [ OK ]
Shutting down loopback interface lo ..... [ OK ]
Bringing up loopback interface lo ..... [ OK ]
Bringing up Ethernet interface eth0 ..... [ OK ]
Set up default gateway ..... [ OK ]
ifconfig
eth0
Link encap:Ethernet HWaddr 50:9A:C2:02:67:00
inet addr:172.50.0.20 Bcast:172.50.255.255 Mask:255.255.0.0
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:171 errors:0 dropped:0 overruns:0 frame:0
TX packets:158 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:53747 (52.1 KiB) TX bytes:54096 (52.8 KiB)

lo
Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)
```

Win-1 10.0.0.10
Win-2 10.0.0.20
Net 10.0.0.1
R1 20
40.0.0.2
R4 200.12.66.1
Net 200.12.66.1
DNS 200.12.66.10

Linux-1 172.50.0.10
Linux-2 172.50.0.20

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

ACL

Rain coming
In about 3 hours

Search

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12:33
20-06-2026

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SUREN NETWORK
PLAY WITH TECHNOLOGIES

Win-1
10.0.0.10

Win-2
10.0.0.20

Net
10.0.0.10

R1
10.0.0.1

R2
20.0.0.1

QEMU (Win-1)

File Edit View History Bookmarks Tools Help

http://172.50.0.10/ http://172.50.0.20/ Problem loading page Problem loading page

172.50.0.10 Search

Hello World!, Hello everyone This file is located in /home/tmpd

R1

IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type
10.0.0.2	0150.f327.027a.00	Jun 21 2026 07:24 AM	Automatic
10.0.0.3	0150.43a8.027b.00	Jun 21 2026 07:25 AM	Automatic

```
R1#  
R1#  
R1#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
R1(config)#access-list 1 permit 10.0.0.2 0.255.255.255  
R1(config)#int e0/0  
R1(config-if)#access-group 1 in  
% Invalid input detected at '^' marker.  
R1(config-if)#ip access-group 1 in  
R1(config-if)#
```

Firefox automatically sends some data to Mozilla so that we can improve your experience. Choose What I Share

Start CHWINDOWS\system32\ Network Connections S Firefox

Physical 01 : 00 : 00 START

Hot weather Now

Search

ENG IN 13:01 20-06-2026

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QEMU (Win-2)

Internet Explorer

Win-1
10.0.0.10

Win-2
10.0.0.20

Net

QEMU (Win-2)

```
C:\WINDOWS\system32\cmd.exe
Reply from 172.50.0.20: bytes=32 time=2ms TTL=61
Reply from 172.50.0.20: bytes=32 time=1ms TTL=61
Reply from 172.50.0.20: bytes=32 time=1ms TTL=61
Reply from 172.50.0.20: bytes=32 time=1ms TTL=61
Ping statistics for 172.50.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>ping 172.50.0.20
Pinging 172.50.0.20 with 32 bytes of data:
Reply from 18.0.0.1: Destination net unreachable.
Reply from 18.0.0.1: Destination net unreachable.
Reply from 18.0.0.1: Destination net unreachable.
Reply from 18.0.0.1: Destination net unreachable.
Ping statistics for 172.50.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Documents and Settings\user>
```

Windows has finished installing new devices. The software that supports your device requires that you restart your computer. You must restart your computer before the new settings will take affect.

Do you want to restart your computer now?

Yes No

Windows - System Error
A duplicate name exists on the network.

OneDrive - Personal
Up to date

01 : 00 : 00 START

34°C Sunny

Search

ENG IN

13:07 20-06-2026

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Notifications

SUREN NETWORK
PLAY WITH TECHNOLOGIES

Win-1
10.0.0.10

Win-2
10.0.0.20

Net

10.0.0.1

R1

20.0.0.1

R1

```
*Jun 20 07:36:39.821: %SYS-5-CONFIG_I: Configured from console by console
R1#sh access-list
Standard IP access list 1
  10 permit 10.0.0.0, wildcard bits 0.255.255.255
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no access-list
% Incomplete command.
R1(config)#no access-list 1
R1(config)#access-list 1 permit 10.0.0.2 0.0.0.0
R1(config)#exit
R1#sh
*Jun 20 07:37:52.661: %SYS-5-CONFIG_I: Configured from console by console
R1#sh access-list
Standard IP access list 1
  10 permit 10.0.0.2
```

40.0.0.2

200.12.66.1

R4

Net

DNS

200.12.66.10

ACL

Physical

01 : 00 : 00 START

34°C Sunny

Search

ENG IN

13:08 20-06-2026

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Notifications

SUREN NETWORK
PLAY WITH TECHNOLOGIES

Win-1
10.0.0.10

Win-2
10.0.0.20

Net

10.0.0.1

R1

20.0.0.1

R1

```
*Jun 20 07:37:52.661: %SYS-5-CONFIG_I: Configured from console by console
R1#sh access-list
Standard IP access list 1
  10 permit 10.0.0.2
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 1 permit 10.0.0.3 0.0.0.0
% Incomplete command.
R1(config)#access-list 1 permit 10.0.0.3 0.0.0.0
R1(config)#exit
R1#
*Jun 20 07:39:21.060: %SYS-5-CONFIG_I: Configured from console by console
R1#sh access-list
Standard IP access list 1
  10 permit 10.0.0.2
  20 permit 10.0.0.3
R1#
```

40.0.0.2

200.12.66.1

R4

Net

DNS

200.12.66.10

ACL

Physical

01 : 00 : 00 START

34°C Sunny

Search

ENG IN

13:08 20-06-2026

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QEMU (Win-2)

Internet Explorer

Win-1 10.0.0.10

Win-2 10.0.0.20

Net

QEMU (Win-2)

```
C:\WINDOWS\system32\cmd.exe
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Ping statistics for 172.50.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Documents and Settings\user>ping 172.50.0.20
Pinging 172.50.0.20 with 32 bytes of data:
Reply from 172.50.0.20: bytes=32 time=2ms TTL=61
Reply from 172.50.0.20: bytes=32 time=2ms TTL=61
Reply from 172.50.0.20: bytes=32 time=1ms TTL=61
Reply from 172.50.0.20: bytes=32 time=1ms TTL=61
Ping statistics for 172.50.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>
```

Windows has finished installing new devices. The software that supports your device requires that you restart your computer. You must restart your computer before the new settings will take affect.

Do you want to restart your computer now?

Yes No

Windows - System Error

A duplicate name exists on the network.

2.56.10

Physical

01 : 00 : 00 START

34°C Sunny

Search

ENG IN

13:09 20-06-2026